Optimal Fusion-Based Target Detection With Multichannel ATI SAR

Min Tian, *Member, IEEE*, and Bin Liao, *Senior Member, IEEE*

***Abstract*— In this article, an optimal fusion-based target detector with multichannel along-track interferometry (ATI) synthetic aperture radar (SAR) is investigated for ground moving target indication (GMTI) in a heterogeneous background. In the proposed method, a group of multibaseline ATI-phase tests for fully exploiting the spatial degrees of freedom (DoF) and the adaptive matched filter (AMF) test are first exploited for local target detection. Then, the global detection is designed based on Chair and Varshney's optimal fusion rule. The optimal local detection thresholds are determined by designing the global optimization in terms of the optimal fusion rule. Therefore, the maximum target detection performance under the constant probability of false alarm (Pfa) can be achieved. To facilitate the implementation, a low-complexity solution to the global optimization and a cognitive detection framework for the proposed detector without the prior target knowledge is developed. Simulation and experimental results of the receiver operator characteristic (ROC) demonstrate its significant advantages in detecting dim targets that have a low signal-to-clutter-plus-noise ratio (SCNR) and small radial velocity in heterogeneous clutter backgrounds.**

***Index Terms*— Dim target detection, heterogeneous background, multichannel along-track interferometry (ATI), optimal fusion.**

I. Introduction

GROUND moving target indication (GMTI) with airborne or space-based synthetic aperture radar (SAR), which generally deploys an antenna array with multiple phase centers and multichannel space–time adaptive processing (STAP) [1], [2], [3], has received growing attention, owing to its great potentials for detecting the dim targets in the endo-clutter region [4], [5]. In particular, SAR with broad bandwidth and long coherent processing time interval (CPI) can achieve high spatial resolution of both ground clutter [6], [7] and moving targets [8], [9], [10], [11], [12] and, hence, can reduce the incident clutter strength and provide abundant clutter sample support for effective ground clutter suppression in homogeneous background [4], [5].

Note that, in certain complex environments such as cities and outskirts, clutter data typically consist of a mixture of backscatter types with variable reflectivity, such as cultivated land, forests, and man-made objects, leading to heterogeneous or even extremely heterogeneous detection backgrounds [13]. Specifically, the clutter amplitude statistically appears as a long tail in the high values, which does not obey the Rayleigh distribution [14]. Generally, the heterogeneous clutter signal can be properly characterized with the product model or multiplicative noise model [14], [15], [16], [17], [18]. In this case, performance degradation of clutter cancellation occurs due to the lack of independent and identically distributed (i.i.d.) clutter samples, and the isolated and large clutter suppression residuals result in a high probability of false alarm (Pfa) [19], [20]. In [21], a two-stage idea is exploited to design tunable detectors capable of ensuring the same target detection performance, as Kelly's detector under perfectly matching conditions, and enhanced rejection capabilities of unwanted signals, such as the two-step detector by cascading Kelly's detector [22] and the KWA [23], and Kelly's detector and the Rao test [24], which can guarantee an appropriate tradeoff between rejection of interferences and detection of targets in the Gaussian clutter background. However, they may not work properly in non-Gaussian background since any test of the two-stage detectors depends on the adaptive clutter suppression result and, thus, inevitably detects many false alarms due to the heterogeneous clutter.

For the abovementioned approaches, increasing the target detection threshold can reduce the false alarm, whereas the weak and slow targets may fail to be detected due to the severe target signal-to-noise losses incurred in the multichannel ground clutter adaptive suppression [4], [5]. In the last decades, along-track interferometry (ATI) techniques [25] have been extensively exploited for slow target detection [15], [16], [17], [18], [26], [27], [28], [29]. From the statistical point of view, the along-track interferometric phase is proportional to the line-of-sight velocity of the backscatter [26], [27], [28], which has been utilized for the discrimination of moving targets in heterogeneous clutter environments [15], [16], [17], [18], [29]. The two-stage detection method by cascading the interferometry magnitude and phase (IMP) in [18] and the constant false-alarm rate (CFAR) detector based on a joint metric of dual-channel SAR IMP in [17] can effectively reduce the false alarms in the heterogeneous background. However, they may fail to detect the targets with a low signal-to-clutter-plus-noise ratio (SCNR) due to their limited clutter suppression ability.

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Min Tian is with Xidian University, Xi'an 710071, China, and also with the Hangzhou Institution of Technology, Xidian University, Hangzhou 312000, China, and also with the School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: tianmin@xidian.edu.cn).

Bin Liao is with the State Key Laboratory of Radio Frequency Heterogeneous Integration and Guangdong Provincial Key Laboratory of Intelligent Information Processing, Shenzhen University, Shenzhen 518060, China (e-mail: binliao@szu.edu.cn).

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For detecting the low-SCNR targets in the heterogeneous background, two-step detectors with strong clutter cancellation capability were proposed in [15] and [16]. In the two-step detectors, the magnitude test based on the multichannel adaptive clutter suppression result is adopted in the first stage. Usually, a low target detection threshold is set to detect the potential targets, while false alarms may occur due to large clutter suppression residuals in the heterogeneous background. In the second stage, the false alarms are removed by using additional tests, such as the degree of radial-velocity consistency (DRVC) test [15] and the ATI-phase test [16]. By doing so, better minimum discernible velocity (MDV) and minimum discernible SCNR of targets can be obtained.

It should be pointed out that the tests in the second stage of the two-step detectors [15], [16] may limit the availability of the spatial degree of freedoms (DoFs) and, therefore, restrict the detection performance. Compared with the standard dual-channel ATI-phase test, there exist many more independent ATI phases through processing ATI on all pairs of dual-channel data in the multichannel case. The ATI phases measured by different baseline lengths can exhibit distinct sensitivities to the backscatter's motions along the radar's line-of-sight direction. In spite of the fact that it is promising to exploit the multiscale ATI-phase tests for SAR GMTI, to the best of our knowledge, there are few discussions in the open literature. To put the idea into practice, a core problem is how to properly utilize and fuse the results. Toward this end, we propose a new ground moving target detector based on optimal fusion with multichannel ATI SAR.

In this work, we propose to expand the dual-channel ATI processing into the multichannel processing case and then formulate a group of multibaseline ATI-phase tests to make full use of the spatial DoF. In the proposed detector, the multibaseline ATI-phase tests and the adaptive matched filter (AMF) are performed individually as local target detection. On this basis, a global test is formulated to fuse these local decisions by the optimal fusion rule. Since the local detection thresholds are optimized in terms of the optimal fusion rule for the max probability of detection (Pd) under the constant Pfa, the globally optimal target detection performance with the multiscale tests can be obtained under the CFAR rule. Furthermore, to facilitate the implementation in practice, a low-complexity solution to the joint optimization process and a cognitive detection framework for the proposed detector without prior information are developed. Extensive simulations and real-data experiments are performed to validate the superiority of the proposed detector in the heterogeneous clutter background.

Notations: $(\cdot)^{\mathsf{T}}$ and $(\cdot)^{\mathsf{H}}$ are the operators of transpose and conjugate transpose, respectively. $*$ is the conjugate operator. $\mathrm{Re}\{\cdot\}$, $\arg[\cdot]$, and $|\cdot|$ denote the real part, phase, and modulus of a complex number, respectively. $E[\cdot]$ is the mathematical expectation. $\mathbf{I}$ is the identity matrix with the appropriate size. i is the imaginary unit with $i^2 = -1$.

II. Mathematical Model

Consider an airborne GMTI SAR with a linear array of N antenna elements uniformly distributed along the aircraft's track direction. The spacing between two adjacent channels is denoted by d, and the radar wavelength is λ. In the radar side-looking work mode, the geometric relationship between the radar and a moving target is shown in Fig. 1(a). For a reference time, the origin o- and the x-axis correspond to the aircraft's projection position on the ground and the track direction (azimuth), respectively. Without loss of generality, the z- and y-axes are, respectively, perpendicular to the ground and the xoz plane. In the $oxyz$ coordinate system, it is assumed that the aircraft moves with a linear velocity V in the direction of x in the CPI. The azimuth and incident angles of the target [e.g., the vehicle in Fig. 1(a)] are expressed by α and θ, respectively. In practice, the aircraft may yaw and roll due to the meteorological factors in the air. As shown in Fig. 1(b) and (c), the yaw and roll angles are expressed by γ and β, respectively, and they can be estimated by the positioning system. In this case, the azimuth and incident angles of the target become $\alpha + \gamma$ and $\theta + \beta$, respectively, and the radial velocity of the target by projecting its speed to the radar line-of-sight direction is denoted by v_{r}.

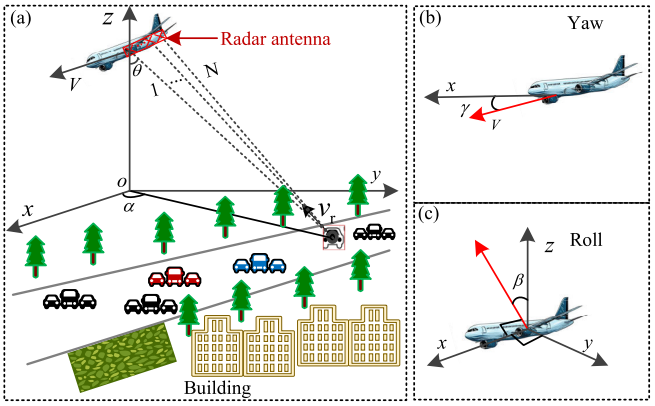


Fig. 1. (a) Geometric relationship between the multichannel radar and a ground moving target. (b) Aircraft yaw. (c) Aircraft roll.

During ground moving target detection, the GMTI radar first emits broad-bandwidth electromagnetic waves over a long CPI via the N channels. Then, identical signal processing chains are implemented to process the sampled data of all airborne radar array element channels over the CPI. After the process, assume that a slowly moving ground target is properly match-filtered over the radar signal bandwidth and coherently integrated over the entire CPI, and N range-Doppler maps in the case of range-Doppler processing or SAR imaging in the range and cross–range domains output from the array channels. Afterward, the spatial position alignment for the ground scatters or moving targets in the N SAR images and the elimination of the constant amplitude and phase differences from each other SAR image due to the fixed array element channel response errors are completed by the image registration and calibration processing [25].

For a pixel $k \in \{1, 2, \ldots, K\}$, denote by $\mathbf{z}(k) = [z_1(k), z_2(k), \ldots, z_N(k)]^{\mathsf{T}}$ a snapshot of $N \times 1$ data vector from the processed N SAR images, where $z_p(k)$ is the complex signal in the pth SAR image, $p \in \{1, 2, \ldots, N\}$. Accordingly, the binary hypothesis test for the purpose of

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target signal detection is defined as

$$H_0 : \mathbf{z}(k) = \mathbf{c}(k) + \mathbf{n}(k)$$
$$H_1 : \mathbf{z}(k) = \mathbf{s}(k) + \mathbf{c}(k) + \mathbf{n}(k) \quad (1)$$

where H_0 is the target-absent hypothesis, while H_1 is the target-present hypothesis; $\mathbf{c}(k)$ and $\mathbf{s}(k)$ are, respectively, the clutter vector and the signal vector of a deterministic moving target, and the noise vector $\mathbf{n}(k)$ obeys a zero-mean complex Gaussian distribution as $\mathbf{n}(k) \sim \mathcal{CN}(\mathbf{0}, \sigma_{\mathrm{n}}\mathbf{I})$ with σ_{n} being the noise power.

For the pixel k, the amplitudes of the focused target signal in all channels are supposed to be identical such that $\mathbf{s}(k)$ can be written as

$$\mathbf{s}(k) = \xi_{\mathrm{s}}(k)\mathbf{a}(k) \quad (2)$$

where $\xi_{\mathrm{s}}(k)$ denotes the complex target amplitude and $\mathbf{a}(k)$ is the spatial steering vector corresponding to the target. Under the radar working mode of full aperture transmission and subaperture reception, the effective phase centers of any two adjacent receiving channels are spaced by $d/2$, which is also referred to as the effective baseline length [30]. For a moving object with the radial velocity $v_{\mathrm{r}}(k)$ in the pixel k, its Doppler shift $f(k) = 2v_{\mathrm{r}}(k)/\lambda$ will result in an angle $2\pi f(k)(d/2/V)$ during the array passing through the effective baseline, where λ refers to the radar wavelength. Accordingly, we have

$$\mathbf{a}(k) = \left[1, \exp\left(\mathrm{i}2\pi f(k)\frac{d}{2V}\right), \ldots, \exp\left(\mathrm{i}2\pi f(k)\frac{(N-1)d}{2V}\right)\right]^{\mathsf{T}}. \quad (3)$$

In the case of broad bandwidth and long CPI, the details of ground scatter can be exhibited clearly in the high-resolution SAR images. In practice, due to the composite scattering including buildings, trees, grass, and roads, as shown in Fig. 1, ground clutter is often statistically nonuniform so that we can use the product model [14], [15], [16], [17], [18] to describe the heterogeneous clutter statistics, i.e.,

$$\mathbf{c}(k) = \Delta(k)\mathbf{c}_0(k) \quad (4)$$

where the texture random variable $\Delta(k) \in [0, \infty)$ describes the amplitude change of the clutter backscatter, the complex Gaussian vector $\mathbf{c}_0(k) \sim \mathcal{CN}(\mathbf{0}, \mathbf{R})$ denotes the common homogeneous clutter signal with the complex zero-mean Gaussian distribution, and $\mathbf{R}$ is the homogeneous clutter covariance matrix. Various statistical models for the texture have been investigated for typical clutter scenes in SAR images [14], [15], [16], [17], [18]. Extensive experimental evidence based on RADARSAT-2's clutter measurements has demonstrated that the inverse chi-square function can appropriately characterize Δ for urban areas [16], [18]

$$f_{\Delta}(\delta) = \frac{2(\chi-1)^{\chi}}{\Gamma(\chi)}\delta^{-(2\chi+1)}\exp\left(-\frac{\chi-1}{\delta^2}\right) \quad (5)$$

where the degree of heterogeneity for the clutter background can be reflected by the texture parameter χ. Specifically, the central moments of the texture Δ is

$$E\left[\Delta^r\right] = (\chi-1)^{r/2}\frac{\Gamma(\chi - r/2)}{\Gamma(\chi)} \quad (6)$$

where r denotes the moment and $\Gamma(\cdot)$ is the Gamma function. The less the χ, the more the heterogeneous of the background.

III. Proposed Method

In this section, we present an optimal fusion-based detector with multichannel ATI SAR. The functional block diagram of this method is shown in Fig. 2. In brief, in the proposed method, multiscale detectors, i.e., the AMF test (T) and multibaseline ATI-phase tests ($\varphi_1, \varphi_2, \ldots, \varphi_M$), are constructed with N SAR images. During this process, the respective distribution characteristics of the multiscale tests under the two hypotheses are first estimated such that the receiver operator characteristics (ROCs) of each local test can be properly predicted. Next, the optimal local detection thresholds $\eta_0, \eta_2, \ldots, \eta_M$ are determined by the global optimization for the max Pd under the CFAR. Finally, the local decisions $u_0, u_1, \ldots, u_M$ are decided, and a global test can be formulated to fuse these decisions via the optimal fusion weights $a_0, a_1, \ldots, a_M$. In what follows, the details of the proposed method are detailed.

A. General Detection Principle

1) Local Detection: Under the assumption of heterogeneous ground clutter and Gaussian noise background, multiscale tests are defined in the local detection process for the purpose of target signal detection. To increase the target SCNR, each pixel in the form of either range-Doppler maps or SAR images is processed by the AMF [31] that optimally combines the array outputs only in the spatial dimension with the $N \times 1$ weight vector $\mathbf{u}$ to reject the strong ground clutter. The AMF is a degenerate case of STAP, which is capable of delivering approximately optimal clutter suppression performance in a long CPI and offers substantial advantages for practical applications, such as reduced algorithmic complexity, faster convergence, and lower computational load. For heterogeneous environments, it is often difficult to acquire enough i.i.d. samples to estimate the clutter plus noise covariance matrix for adaptive clutter rejection processing. Fortunately, plenty of methods have been proposed for designing the adaptive clutter rejection weight $\mathbf{u}$ in recent decades [32], [33], [34], [35], [36], [37], [38]. Consider that a slowly ground moving target may appear at several spatially continuous pixels after the adaptive clutter rejection, and the multilook processing is utilized to further increase the target-to-background contrast. As such, according to [16] and [39], statistically independent cells are selected around the test pixel to formulate the multilook AMF detector via the average operation

$$T = \sum_{k=1}^{n}\frac{\mathbf{u}^{\mathsf{H}}(k)\mathbf{z}(k)\mathbf{z}^{\mathsf{H}}(k)\mathbf{u}(k)}{\sigma_{\mathrm{cn}}} \quad (7)$$

where n is the look number and $\mathbf{u}^{\mathsf{H}}(k)\mathbf{u}(k) = 1$, while σ_{cn} is the average power of the adaptive clutter suppression residuals.

Meanwhile, a set of extending ATI phase tests are obtained independently for the local decisions on whether or not a moving target signal appears in the SAR image pixel. For the multichannel ATI SAR, if a dual-channel baseline is composed of the receiving channels l and m, the n-look

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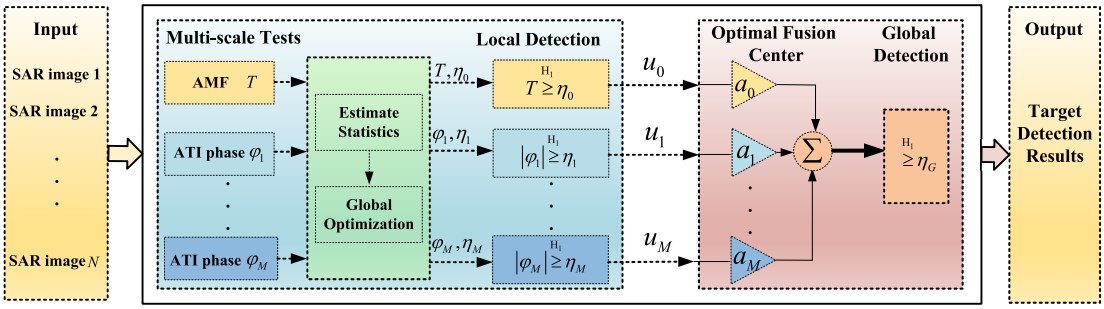


Fig. 2. Functional block diagram of the proposed detection method.

interferometric phase in terms of the effective baseline length $d_{l,m} = (m-l)d/2$ is measured by [26], [27], [28]

$$\varphi(l,m) = \arg\left[\sum_{k=1}^{n} z_l(k) z_m^*(k)\right] \tag{8}$$

which can be approximated as

$$\varphi(l,m) \approx \left[2\pi \frac{2v_{\mathrm{r}}}{\lambda}\frac{d_{l,m}}{V}\right]_{\pi} \in [-\pi, \pi] \tag{9}$$

for a moving object with a radial velocity v_{r} in the pixel under the test [18]. In (9), $[\cdot]_{\pi}$ represents the phase wrapping in the cycle of 2π. In principle, $\varphi(l,m)$ measures the scatterer's motions along the radar line-of-sight direction by the signal phase difference between the two channels. Generally, $\varphi(l,m) \approx 0$ for the stationary ground clutter, while $\varphi(l,m) \neq 0$ for moving targets in the case of unambiguous phase. This provides distinguishable signatures for detecting the moving targets from the ground background. To make better use of the spatial DoF in the N-channel ATI SAR, the complex n-look ATI processing is applied to all dual-channel data to form $M = N(N-1)/2$ ATI-phase tests via (8)

$$\begin{aligned} \varphi_1 &= \varphi(1,2), \ldots, \varphi_{N-1} = \varphi(1,N) \\ \varphi_N &= \varphi(2,3), \ldots, \varphi_{2N-3} = \varphi(2,N) \\ &\vdots \\ \varphi_{N(N-1)/2} &= \varphi(N-1,N). \end{aligned} \tag{10}$$

To ensure effective target detection and tracking processing in the radar receiver, the detection Pfa is usually fixed in advance, and then, the detection threshold is determined with the constant Pfa and the statistics of the clutter-plus-noise background, which is referred to as the CFAR detection rule [40]. Thus, the local detection Pfa should be carefully set to achieve the constant global Pfa and good detection result. The determination approach will be presented in Section III-B.

Assume that the optimal local detection Pfas have been determined, i.e., P_{f0} for the AMF test and $P_{\mathrm{f1}}, P_{\mathrm{f2}}, \ldots, P_{\mathrm{f}M}$ for M ATI-phase tests, which can ensure the maximum target detection performance under the given global Pfa. In the AMF detection, the hypothesis H_1 is declared for the pixel under the test if T exceeds the threshold η_0

$$T \geq \eta_0 \tag{11}$$

where η_0 and P_{f0} satisfy the following condition:

$$P_{\mathrm{f0}} = \int_{\eta_0}^{+\infty} f_{\mathrm{AMF}}(t, \hat{n}, \chi; H_0)\, dt \tag{12}$$

and $f_{\mathrm{AMF}}(t, \hat{n}, \chi; H_0)$ is the probability distribution function (pdf) of the AMF test under the hypothesis H_0, which is characterized by [16], [39]

$$f_{\mathrm{AMF}}(t, \hat{n}, \chi; H_0) = \frac{\hat{n}^{\hat{n}} t^{\hat{n}-1}}{\Gamma(\hat{n})} \frac{\Gamma(\hat{n}+\chi)}{\Gamma(\chi)} \frac{(\chi-1)^{\hat{n}+\chi-1}}{(\chi-1+\hat{n}t)^{\hat{n}+\chi}} \tag{13}$$

where the effective look number $\hat{n}$ is directly acquired by $\hat{n} \approx E[T]$ and the texture parameter is estimated via the second moment of the AMF test data as

$$\hat{\chi} = \frac{2\hat{n}m_2 - (\hat{n}+1)}{\hat{n}m_2 - (\hat{n}+1)} \tag{14}$$

where $m_2 = E[|T|^2]$. With the statistics estimation results, η_0 can be determined for a given P_{f0} via (12).

Supposing the AMF test for a deterministic moving target, i.e., $T = \omega$, the pdf under the hypothesis H_1 becomes

$$\begin{aligned} &f_{\mathrm{AMF}}(t, \hat{n}, \chi, \omega; H_1) \\ &= \frac{\hat{n}^{\hat{n}}\Gamma(\hat{n}+\chi)}{\Gamma(\hat{n})\Gamma(\chi)} \frac{(\chi-1)^{\chi} t^{\hat{n}-1}}{(\omega+\chi-1+\hat{n}t)^{\hat{n}+\chi}} \\ &\quad {}_2F_1\left(\frac{\hat{n}+\chi}{2}, \frac{\hat{n}+\chi+1}{2}; \hat{n}; \frac{4\hat{n}\omega t}{(\omega+\chi-1+\hat{n}t)^2}\right) \end{aligned} \tag{15}$$

where ${}_2F_1(\cdot)$ is the Gauss' hypergeometric function. Accordingly, the local Pd can be computed via the following integral:

$$P_{\mathrm{d0}} = \int_{\eta_0}^{+\infty} f_{\mathrm{AMF}}(t, \hat{n}, \chi, \omega; H_1)\, \mathrm{d}t. \tag{16}$$

For the local detection with the ATI-phase test φ_j, $1 \leq j \leq M$, the hypothesis H_1 is declared when the absolute value of the test exceeds the detection threshold η_j, i.e.,

$$|\varphi_j| \geq \eta_j. \tag{17}$$

With the statistical modeling results in [26], [27], and [18], the pdf for the test φ_j under the hypothesis H_0 is formulated by

$$\begin{aligned} &f_{\mathrm{p}}(\varphi_j, \hat{n}, \rho_j, \varphi_{\mathrm{c}j}; H_0) \\ &= \frac{\Gamma(\hat{n}+1/2)(1-\rho_j{}^2)^{\hat{n}} \rho_j \cos(\varphi_j - \varphi_{\mathrm{c}j})}{2\sqrt{\pi}\Gamma(\hat{n})(1-\rho_j{}^2\cos^2(\varphi_j - \varphi_{\mathrm{c}j}))^{\hat{n}+1/2}} \end{aligned}$$

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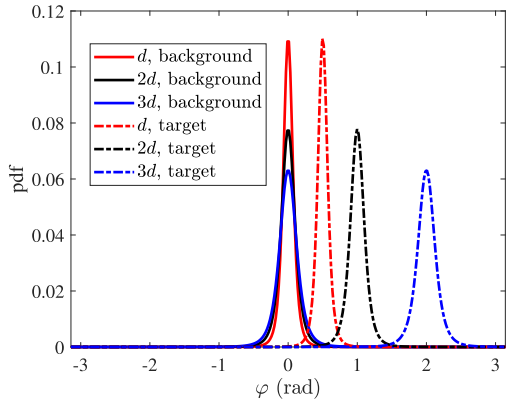


Fig. 3. Theoretical statistics comparisons for the ATI-phase tests under the two hypotheses on three effective baselines d, $2d$, and $3d$.

$$+\frac{\left(1-{\rho_j}^2\right)^{\hat{n}}}{2\pi}{}_2F_1\left(\hat{n},1;1/2;{\rho_j}^2\cos^2\left(\varphi_j-\varphi_{cj}\right)\right) \quad (18)$$

where φ_{cj} and ρ_j denote the ATI-phase estimate and the channel correlation coefficient for the clutter background, respectively. φ_{cj} is commonly set to zero for land applications. The correlation coefficient between the channels l and m is estimated over all samples in the dataset as

$$\hat{\rho}_j=\frac{\left|E\left[z_l z_m^*\right]\right|}{\sqrt{E\left[|z_l|^2\right]E\left[|z_m|^2\right]}} \quad (19)$$

where a longer effective baseline length would lead to a smaller $\hat{\rho}_j$, owing to more signal decorrelation.

Assume that the target parameters φ_{tj} and ρ_{tj} are known or estimated based on target samples. Then, the ATI-phase test pdf under the hypothesis H_1 is [18], [26], [27]

$$\begin{aligned}&f_p\left(\varphi_j,\hat{n},\rho_{tj},\varphi_{tj};H_1\right)\\&=\frac{\Gamma\left(\hat{n}+1/2\right)\left(1-{\rho_{tj}}^2\right)^{\hat{n}}\rho_{tj}\cos\left(\varphi_j-\varphi_{tj}\right)}{2\sqrt{\pi}\Gamma\left(\hat{n}\right)\left(1-{\rho_{tj}}^2\cos^2\left(\varphi_j-\varphi_{tj}\right)\right)^{\hat{n}+1/2}}\\&\quad+\frac{\left(1-{\rho_{tj}}^2\right)^{\hat{n}}}{2\pi}{}_2F_1\left(\hat{n},1;1/2;{\rho_{tj}}^2\cos^2\left(\varphi_j-\varphi_{tj}\right)\right).\end{aligned} \quad (20)$$

Through estimating each ATI-phase test statistics under the hypothesis H_0 by (18) and (19), the local detection thresholds $\eta_1,\eta_2,\ldots,\eta_M$ can be separately obtained via

$$\begin{aligned}P_{fj}&=P\left(|\varphi_j|\geq\eta_j;H_0\right)\\&=\int_{\eta_j}^{\pi}f_p\left(\varphi_j,\hat{n},\rho_j,\varphi_{cj};H_0\right)d\varphi_j+\int_{-\pi}^{-\eta_j}f_p\left(\varphi_j,\hat{n},\rho_j,\varphi_{cj};H_0\right)d\varphi_j.\end{aligned} \quad (21)$$

Based on the statistics under the hypothesis H_1 in (20), the local Pd in accordance with the threshold η_j is given by

$$\begin{aligned}P_{dj}&=P\left(|\varphi_j|\geq\eta_j;H_1\right)\\&=\int_{\eta_j}^{\pi}f_p\left(\varphi_j,\hat{n},\rho_{tj},\varphi_{tj};H_1\right)d\varphi_j+\int_{-\pi}^{-\eta_j}f_p\left(\varphi_j,\hat{n},\rho_{tj},\varphi_{tj};H_1\right)d\varphi_j.\end{aligned} \quad (22)$$

From the perspective of statistics, the ATI-phase tests distribute independently of each other and, in general, do not depend on the heterogeneity degree of the clutter background. Besides, the test measured by a longer effective baseline would more sensitively reflect the moving object's motions at the radar line-of-sight direction, as expressed in (9). A direct theoretical statistics comparison at $\hat{n}=2$ with respect to different effective baselines is shown in Fig. 3. For the ATI-phase tests of the effective baselines d, $2d$, and $3d$, the channel correlation coefficients are 0.99, 0.98, and 0.97, respectively, and the target ATI-phase estimates are 0.5, 1, and 1.5 rad, respectively. The curves clearly illustrate the distribution changes at different parameters and more statistical differences between the moving target and clutter background at a longer effective baseline.

After performing the above local detection under the given local Pfa individually, these local tests yield $M+1$ binary decisions $u_0,u_1,\ldots,u_M$ for each cell under the test, where $u_j=+1$ if the hypothesis H_1 is declared and $u_j=-1$ in the otherwise, $0\leq j\leq M$, and each local Pd can be predicted, i.e., $P_{d0},P_{d1},\ldots,P_{dM}$.

2) Global Detection: Through the local detection, the dissimilar, statistically independent and multiscale signatures captured in the local tests are decided as a group of binary decisions u_j on whether or not a moving target presents. For the final decision on the pixel under the test, the global test is generally formulated by

$$Y=a_0u_0+a_1u_1+a_2u_2+\cdots+a_Mu_M \quad (23)$$

where $a_0,a_1,\ldots,a_M$ are the fusion weights.

Most existing two-step detection methods [15], [16], [18] employ the fusion rule that the presence of a target signal is declared only if both local detection stages are above the respective thresholds. From this, their global detection is

$$Y=u_1+u_2\underset{H_0}{\overset{H_1}{\geq}}2. \quad (24)$$

Note that the "and" fusion rule in (24) may not acquire the optimal target detection performance for two or more independent and unequal observations. In fact, for the optimal data fusion strategy, the typical rules "and" and "or" have been discussed under the exponentially distributed observations [41], but the conclusions about the optimal choice cannot suit general cases. In [42], an optimal fusion rule based on the likelihood ratio test using the minimum cost criterion was developed by Chair and Varshney. Once the local detection decisions (or observations) and the corresponding reliability are deterministic, Chair and Varshney's optimal fusion rule is able to acquire the maximum target detection performance.

For the proposed method, the global detection based on Chair and Varshney's optimal fusion rule [42] is derived as

$$\frac{P(u_0,\ldots,u_M;H_1)}{P(u_0,\ldots,u_M;H_0)}\underset{H_0}{\overset{H_1}{\geq}}\frac{P(H_0)(J(0,H_1)-J(1,H_1))}{P(H_1)(J(1,H_0)-J(0,H_0))} \quad (25)$$

with

$$J(1,H_0)>J(0,H_0) \quad (26)$$

where $P(H_1)$ and $P(H_0)$ are the prior probabilities for the hypotheses H_1 and H_0, respectively, and $J(p,H_q)$ is the error

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cost function for detector choosing p when H_q is true [43]. For example,

$$J(0, H_0) = J(1, H_1) = 0 \tag{27a}$$

$$J(1, H_0) = J(0, H_1) = t > 0 \tag{27b}$$

which indicates more costly for the error judgment compared with the correct one.

Transforming (25) with the log function and following the assumptions in (27), we have

$$\log \frac{P(u_1, \ldots, u_M; H_1)}{P(u_1, \ldots, u_M; H_0)} \geq \eta_{\mathrm{G}} \tag{28}$$

where $\eta_{\mathrm{G}} = \log(P(H_0)/P(H_1))$ is the global detection threshold, such as $\eta_{\mathrm{G}} = 0$ if $P(H_0) = P(H_1)$.

To proceed, each local decision is decided individually so that the joint probabilities of the local decisions under the two hypotheses can be simplified as

$$P(u_0, \ldots, u_M; H_1) = \prod_{j=0,1,\ldots,M} P\left(u_j; H_1\right) \tag{29a}$$

$$P(u_0, \ldots, u_M; H_0) = \prod_{j=0,1,\ldots,M} P\left(u_j; H_0\right). \tag{29b}$$

In the local decisions $u_0, u_1, \ldots, u_M$, denote by S_1 and S_2 the index sets of $u_j = +1$ and $u_k = -1$, respectively, and then, the global test in (29) can be recast as

$$\log \frac{P(u_0, \ldots, u_M; H_1)}{P(u_0, \ldots, u_M; H_0)} = \log\left(\prod_{j\in S_1} \frac{P\left(u_j = +1; H_1\right)}{P\left(u_j = +1; H_0\right)} \prod_{k\in S_2} \frac{P(u_k = -1; H_1)}{P(u_k = -1; H_0)}\right). \tag{30}$$

Let $P(u_j = +1; H_1) = P_{\mathrm{d}j}$ and $P(u_j = +1; H_0) = P_{\mathrm{f}j}$, which are, respectively, the Pd and Pfa for the local detection. One gets $P(u_j = -1; H_1) = 1 - P_{\mathrm{d}j}$ and $P(u_j = -1; H_0) = 1 - P_{\mathrm{f}j}$. Then, (30) is simplified as

$$\log \frac{P(u_0, \ldots, u_M; H_1)}{P(u_0, \ldots, u_M; H_0)} = \sum_{j\in S_1} \log \frac{P_{\mathrm{d}j}}{P_{\mathrm{f}j}} + \sum_{k\in S_2} \log \frac{1 - P_{\mathrm{d}k}}{1 - P_{\mathrm{f}k}}. \tag{31}$$

By defining the following weights:

$$a_j = \begin{cases} \log \dfrac{P_{\mathrm{d}j}}{P_{\mathrm{f}j}}, & \text{if } u_j = +1 \\ \log \dfrac{1 - P_{\mathrm{f}j}}{1 - P_{\mathrm{d}j}}, & \text{if } u_j = -1 \end{cases} \tag{32}$$

the global test in (31) becomes $\sum_{j=0}^{M} a_j u_j$. Consequently, the global detection is formulated as

$$\sum_{j=0}^{M} a_j u_j \underset{H_0}{\overset{H_1}{\geq}} \eta_{\mathrm{G}} \tag{33}$$

where $a_0, a_1, \ldots, a_M$ are the optimal fusion weights and depend on the local detection reliability.

A schematic for the optimal weights a_j versus $P_{\mathrm{f}j}$ and $P_{\mathrm{m}j} = 1 - P_{\mathrm{d}j}$ [probability of miss (Pm)] is shown in Fig. 4, where Fig. 4(a) and (b) shows the cases of $u_j = +1$ and $u_j = -1$, respectively. A high detection reliability for local decisions would occur when the Pfa and Pm are both low. It is seen that the optimal fusion weights increase with the higher detection reliability. Thus, the optimal fusion rule can smartly fuse these decisions.

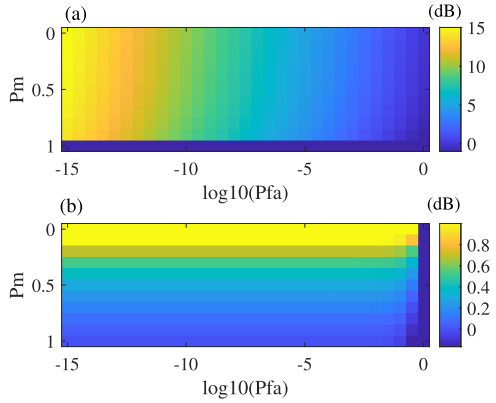


Fig. 4. Optimal fusion weight a_j versus different local detection Pfa ($P_{\mathrm{f}j}$) and Pm ($P_{\mathrm{m}j}$) at two decisions (a) $u_j = +1$ and (b) $u_j = -1$.

B. Global Optimization For the Max Pd Under CFAR

If a group of local detection Pfas ($P_{\mathrm{f}0}, P_{\mathrm{f}1}, \ldots, P_{\mathrm{f}M}$) are fixed, the local detection thresholds ($\eta_0, \eta_1, \ldots, \eta_M$), local decisions ($u_0, u_1, \ldots, u_M$), and local Pds ($P_{\mathrm{d}0}, P_{\mathrm{d}1}, \ldots, P_{\mathrm{d}M}$) are deterministic for the test data. Then, through the optimal fusion rule in (32) and (33), the global target detection design is complete. However, an arbitrary choice of the local detection Pfa cannot yield the global decision, which achieves the max Pd under the CFAR for the proposed detector. To maximize the target detection performance, the global optimization of the proposed detector is designed to determine the optimal local detection Pfa.

1) Optimization Problem: Consider that the optimal local detection Pfa should provide the available maximum Pd under the CFAR for the global detection in terms of the optimal fusion rule, and the global optimization problem is formulated as

$$\max_{P_{\mathrm{f}j}, j=1,\ldots,M} P_{\mathrm{d}} \quad \text{s.t.} \begin{cases} P_{\mathrm{f}} = r \\ 0 \leq P_{\mathrm{f}j} \leq 1 \end{cases} \tag{34}$$

where P_{d} and P_{f} are the global Pd and Pfa, respectively, and r is a fixed global Pfa, such as 10^{-6}.

To find the solution to (34), we first consider a candidate ($P_{\mathrm{f}0}, P_{\mathrm{f}1}, \ldots, P_{\mathrm{f}M}$) for the local detection Pfas, and the local Pds can be determined accordingly, i.e., $P_{\mathrm{d}0}, P_{\mathrm{d}1}, \ldots, P_{\mathrm{d}M}$. Since each local decision has two sides, i.e., $u_j = +1$ or $u_j = -1$, there are totally 2^{M+1} cases via the ordered combination of the $M+1$ local decisions, as shown in Fig. 5. Then, we test each case with the global detection based on (32) and (33). Through the processing, the number of cases that declare H_1 is recorded as L. For any case $l \in \{1, 2, \ldots, L\}$, denote by $\mathbf{D}_{1,l}$ and $\mathbf{D}_{0,l}$ the index sets of the local decisions $+1$ and -1, respectively. As such, the overall Pfa is computed under the

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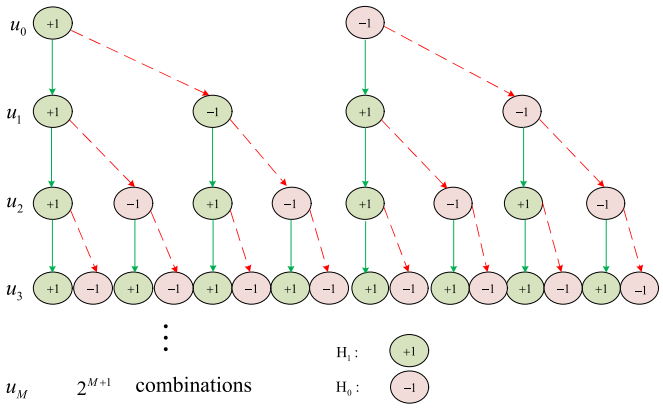


Fig. 5. 2^{M+1} combinations due to $M+1$ local detection for the global test.

hypothesis H_0 by the following expression:

$$P_{\mathrm{f}} = \sum_{l=1}^{L} \prod_{j\in\mathbf{D}_{1,l}} P\left(u_j = +1, H_0\right) \prod_{k\in\mathbf{D}_{0,l}} P(u_k = -1, H_0)$$
$$= \sum_{l=1}^{L} \prod_{j\in\mathbf{D}_{1,l}} P_{\mathrm{f}j} \prod_{k\in\mathbf{D}_{0,l}} (1 - P_{\mathrm{f}k}). \tag{35}$$

In the hypothesis H_1, we compute the overall Pd of the proposed detector as

$$P_{\mathrm{d}} = \sum_{l=1}^{L} \prod_{j\in\mathbf{D}_{1,l}} P\left(u_j = +1, H_1\right) \prod_{k\in\mathbf{D}_{0,l}} P(u_k = -1, H_1)$$
$$= \sum_{l=1}^{L} \prod_{j\in\mathbf{D}_{1,l}} P_{\mathrm{d}j} \prod_{k\in\mathbf{D}_{0,l}} (1 - P_{\mathrm{d}k}). \tag{36}$$

Repeating the above procedures for each candidate in the search region and recording the overall Pfa and Pd, we can get the optimal local detection Pfa for (34).

In theory, the global optimization to determine the optimal local detection Pfas should operate on each pixel of the SAR image as a pixel may be a potential target, characterized with particular statistics under the hypothesis H_1. In practice, we conduct the global optimization for several certain cases/pixels based on the following cognitive detection in Section IV.

2) Low-Complexity Solution: The numerical calculation solution to (34) would become computationally difficult for a large number of local tests, i.e., $M = (N-1)N/2 \gg 1$. For the joint optimization of the optimal fusion rule in [42] and the local detection rule proposed in [43], equal local thresholds have been adopted to alleviate the computational complexity [44]. However, it may be unsuitable for the local tests of different ROCs [45].

For the proposed local detection, it is worth noting that the ATI-phase tests measured by equal effective baseline length can have i.i.d. statistics observed from (18) and (20), leading to the same ROC. Based on this feature, a solution, that is, only i.i.d. local tests share the same local detection Pfa and, thus, have equal local detection threshold, may properly trade off the intended target detection performance against computational complexity. To this end, the multibaseline ATI-phase tests in the proposed method are first divided into subgroups based on the effective baseline length. As shown in Table I, for the N-channel case, there are $N-1$ subgroups $(\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_{N-1})$ with respect to the available effective baseline lengths $(d/2, d, \ldots, (N-1)d/2)$, and a subgroup $\mathbf{A}_j$ has $N-j$ element tests with i.i.d. statistical characteristics.

TABLE I
SUBGROUPS OF THE ATI-PHASE TESTS

Subgroup	$\mathbf{A}_1$	$\mathbf{A}_2$	$\cdots$	$\mathbf{A}_{N-1}$
Effective Baseline Length	$d/2$	d	$\cdots$	$(N-1)d/2$
Element Number	$N-1$	$N-2$	$\cdots$	1

In the subgroup $\mathbf{A}_1$, set equal local detection Pfa, i.e., P_{fA_1}, for each element test. The local Pd and the optimal fusion weight can be identical for each element test, such as P_{dA_1} and

$$a^{+}_{\mathrm{A}_1} = \log \frac{P_{\mathrm{dA}_1}}{P_{\mathrm{fA}_1}} \tag{37a}$$
$$a^{-}_{\mathrm{A}_1} = \log \frac{1 - P_{\mathrm{fA}_1}}{1 - P_{\mathrm{dA}_1}} \tag{37b}$$

which correspond to the local decisions $+1$ and -1, respectively.

During the local detection process, assume that any p_1 local tests in $\mathbf{A}_1$ are decided as $+1$, and the remaining $N-1-p_1$ local tests are decided as -1. Then, the contribute of $\mathbf{A}_1$ to the global test can be computed via

$$b_{1,p_1} = \binom{N-1}{p_1} \left(p_1 a^{+}_{\mathrm{A}_1} - (N-1-p_1) a^{-}_{\mathrm{A}_1}\right) \tag{38}$$

where $0 \le p_1 \le N-1$, and

$$\binom{N-1}{p_1} = \frac{(N-1)!}{p_1!(N-p_1-1)!}. \tag{39}$$

The probabilities of b_{1,p_1} under the two hypotheses can be, respectively, expressed by

$$P\left(b_{1,p_1}; H_0\right) = \binom{N-1}{p_1} P^{p_1}_{\mathrm{fA}_1} \left(1 - P_{\mathrm{fA}_1}\right)^{(N-1-p_1)} \tag{40a}$$
$$P\left(b_{1,p_1}; H_1\right) = \binom{N-1}{p_1} P^{p_1}_{\mathrm{dA}_1} \left(1 - P_{\mathrm{dA}_1}\right)^{(N-1-p_1)}. \tag{40b}$$

From this, for arbitrary subgroup A_j with any p_j local decisions $+1$, the corresponding contribute to the global test is $p_j a^{+}_{\mathrm{A}_j}$, and that of $N-j-p_j$ local decisions at -1 is $-(N-j-p_j)a^{-}_{\mathrm{A}_j}$. Then, the contribute of A_j to the global test is formulated as

$$b_{j,p_j} = \binom{N-j}{p_j} \left(p_j a^{+}_{\mathrm{A}_j} - \left(N-j-p_j\right) a^{-}_{\mathrm{A}_j}\right) \tag{41a}$$
$$\binom{N-j}{p_j} = \frac{(N-j)!}{p_j!\left(N-p_j-1\right)!} \tag{41b}$$
$$a^{+}_{\mathrm{A}_j} = \log \frac{P_{\mathrm{dA}_j}}{P_{\mathrm{fA}_j}} a^{-}_{\mathrm{A}_j} = \log \frac{1 - P_{\mathrm{fA}_j}}{1 - P_{\mathrm{dA}_j}} \tag{41c}$$

where $1 \le j \le N-1$ and $0 \le p_j \le N-j$; $a^{+}_{\mathrm{A}_j}$ and $a^{-}_{\mathrm{A}_j}$, respectively, denote the optimal fusion weights for the local decisions $+1$ and -1.

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Similarly, the probabilities of b_{j,p_j} under the two hypotheses are given by

$$P\left(b_{j,p_j};H_0\right)=\binom{N-j}{p_j}P_{\mathrm{fA}_j}^{p_j}\left(1-P_{\mathrm{fA}_j}\right)^{(N-j-p_j)} \tag{42a}$$

$$P\left(b_{j,p_j};H_1\right)=\binom{N-j}{p_j}P_{\mathrm{dA}_j}^{p_j}\left(1-P_{\mathrm{dA}_j}\right)^{(N-j-p_j)}. \tag{42b}$$

Using $b_0=a_0u_0$ for the contribute of the AMF test to the global test in terms of the optimal fusion weight a_0 and the local decision u_0, then N new tests $b_0, b_{1,p_1},\ldots,b_{N-1,p_{N-1}}$ are formed. Finally, the global detection is recast as

$$Y=b_0+b_{1,p_1}+\cdots+b_{N-1,p_{N-1}}\underset{H_0}{\overset{H_1}{\gtrless}}\eta_{\mathrm{G}}. \tag{43}$$

Under the given detection Pfas for the new tests, in (43), b_0 has two sides, i.e., $b_0=a_0$ if $u_0=+1$ and $b_0=-a_0$ if $u_0=-1$, while b_j, p_j $(1<j<N-1)$ has $N-j+1$ possible values as $0\le p_j\le N-j$ according to (41). Thus, the total number of combinations of the new tests is $2N!$ cases. In the following, decide the global detection in (43) with each case, and then, the amount of cases that are finally declared as the hypothesis H_1 is recorded as L_0. The overall Pfa and Pd of the proposed method can be expressed by

$$P_{\mathrm{f}}=\sum_{l=1}^{L_0}P\left(b_0,b_{1,p_1},\ldots,b_{N-1,p_{N-1}};H_0,l\right) \tag{44a}$$

$$P_{\mathrm{d}}=\sum_{l=1}^{L_0}P\left(b_0,b_{1,p_1},\ldots,b_{N-1,p_{N-1}};H_1,l\right). \tag{44b}$$

As each new test is composed of statistically independent tests, these new tests are also independent of each other in statistics. Therefore, the probabilities of a case l under the two hypotheses in (44) can be expressed by

$$\begin{aligned}&P\left(b_0,b_{1,p_1},\ldots,b_{N-1,p_{N-1}};H_0,l\right)\\&=P(b_0;H_0,l)\times P\left(b_{1,p_1};H_0,l\right)\cdots P\left(b_{N-1,p_{N-1}};H_0,l\right)\end{aligned} \tag{45a}$$

$$\begin{aligned}&P\left(b_0,b_{1,p_1},\ldots,b_{N-1,p_{N-1}};H_1,l\right)\\&=P(b_0;H_1,l)\times P\left(b_{1,p_1};H_1,l\right)\cdots P\left(b_{N-1,p_{N-1}};H_1,l\right).\end{aligned} \tag{45b}$$

As a result, the problem in (34) can be rewritten as

$$\max_{P_{\mathrm{fA}_j},j=0,\ldots,N-1}\ P_{\mathrm{d}}\quad \text{s.t.}\begin{cases}P_{\mathrm{f}}=r\\0\le P_{\mathrm{fA}_j}\le 1.\end{cases} \tag{46}$$

The new global optimization problem in (46) offers substantial advantage in reducing the computational complexity due to the reduced number of N new local tests from that of $(N-1)N/2+1$ original local tests.

Assume that the search number for Pfa is k_0 for each local test. Specifically, given a search range for the Pfa in local detection, for instance, $k_0=31$ and the search region is represented by the set $\mathbf{B}=\{10^{-6},10^{-5.8},10^{-5.6},\ldots,10^{-0.2},1\}$ with 31 elements in ascending order, the candidate number for the local Pfa is reduced to 31^N for the new tests compared with that of $31^{N(N-1)/2+1}$ in the original case. Recording the candidates $\mathbf{C}(l), l\in 1,2,\ldots,31^N$, the following procedures to determine the optimal local detection Pfa are given by Algorithm 1. Note that the clutter and target parameters are estimated from the test data and, respectively, written as

$$\hat{\Theta}_{\mathrm{c}}=\left\{\hat{\chi},\hat{\varphi}_{\mathrm{c}1},\ldots,\hat{\varphi}_{\mathrm{c}M},\hat{\rho}_1,\ldots,\hat{\rho}_M,\hat{n}\right\} \tag{47a}$$

$$\hat{\Theta}_{\mathrm{t}}=\left\{\hat{\omega},\hat{\varphi}_{\mathrm{t}1},\ldots,\hat{\varphi}_{\mathrm{t}M},\hat{\rho}_{\mathrm{t}1},\ldots,\hat{\rho}_{\mathrm{t}M}\right\}. \tag{47b}$$

For each candidate $\mathbf{C}(l)$, the potential cases of the local decision combination that is further used to decide the global detection become $2N!$ from $2^{N(N-1)/2+1}$. Therefore, compared with the original optimization at a computational complexity of $\mathcal{O}(k_0^{N(N-1)/2+1}\times 2^{N(N-1)/2+1})$, the new optimization method, with a computational complexity of $\mathcal{O}(k_0^N\times 2N!)$, is more efficient.

Algorithm 1 Low-Complexity Algorithm for the Global Optimization

Input: $\hat{\Theta}_{\mathrm{c}}$, $\hat{\Theta}_{\mathrm{t}}$, η_{G}, r, $\mathbf{B}$, $\mathbf{C}(l), l=1,2,3,4,\ldots,31^N$..
Output: $\{P_{\mathrm{fA}_0},P_{\mathrm{fA}_1},\cdots,P_{\mathrm{fA}_{N-1}}\}$

1: **Initialization:** $l=1$;
2: Set $\{P_{\mathrm{fA}_0},P_{\mathrm{fA}_1},\cdots,P_{\mathrm{fA}_{N-1}}\}=\mathbf{C}(l)$;
3: Compute $\eta_0,\cdots,\eta_M$ by (12) and (21);
4: Compute $P_{\mathrm{d}0},\cdots,P_{\mathrm{d}M}$ by (16) and (22);
5: Obtain $P_{\mathrm{dA}_1},\cdots,P_{\mathrm{dA}_N}$ based on the subgroups;
6: Compute b_{j,p_j} by (41a) to (41c), $j=1,2,\ldots,N-1$;
7: Compute the probabilities for b_{j,p_j} by (42a) and (42b) for $j=1,2,\ldots,N-1$;
8: Record $2N!$ global tests and the decisions on (43);
9: Compute the probabilities for the $2N!$ global tests by (45a) and (45b);
10: Compute the overall Pfa and Pd by (44a) and (44b);
11: Judge the current Pfa:
    If the Pfa approximates r, record the l and Pd;
    Otherwise, discard the Pd;
12: Set $l=l+1$;
13: Repeat steps 2) to 13) untill $l=31^N$;
14: Find the maximum Pd and the corresponding index l_0;
15: Get Optimal local Pfa $\{P_{\mathrm{fA}_0},\cdots,P_{\mathrm{fA}_{N-1}}\}=\mathbf{C}(l_0)$.

IV. Target Detection Without Prior Knowledge

In the proposed detector, accurate target information in the data needs to be known before the detection, so as to predict the local test statistics under the hypothesis H_1, which is often unavailable in practice. To deal with environmental uncertainties, Haykin has set up the cognitive dynamic system to learn the essential prior knowledge from continuous experiential interactions with the environment [46], [47]. Inspired by the idea, a simple and effective cognitive detection framework is designed to implement the proposed detector, as shown in Fig. 6. In the framework, the proposed detector is conducted twice. In the autosensing stage, the essential target parameters are initialized, and a relatively high overall detection Pfa is set, which aims to preselect the potential targets in the environment as possible. In the enhanced detection, the proposed method is reperformed on each potential target with the feedback information from the environment in order to remove the false alarms. The particular is designed as follows.

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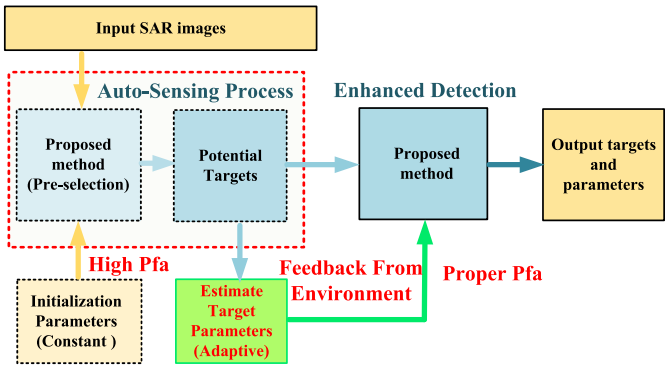


Fig. 6. Cognitive detection framework.

In the autosensing process, the same target parameter set Θ_{t} is first initialized for each pixel. Assume that the MDV v_{m} and the minimum discernible SCNR out of the array element channel, i.e., $k_{\mathrm{scnr}} = \sigma_{\mathrm{s}}/\sigma_{\mathrm{cn}}$, are predicted based on the data and the radar system configuration [5], where σ_{s} is the minimum discernible signal power. The target ATI-phase estimate $\hat{\varphi}_{\mathrm{t}j}(l,m)$ and spatial steering vector $\hat{\mathbf{a}}$ are, respectively, initialized as

$$\hat{\varphi}_{\mathrm{t}j}(l,m) = \left[2\pi \frac{2v_{\mathrm{m}}}{\lambda}\frac{d_{l,m}}{V}\right]_{\pi} \tag{48a}$$

$$\hat{\mathbf{a}} = \left[1, \exp\left(\mathrm{i}\hat{\varphi}_{\mathrm{t}j}(1,2)\right), \ldots, \exp\left(\mathrm{i}\hat{\varphi}_{\mathrm{t}j}(1,M)\right)\right]^{\mathsf{T}}. \tag{48b}$$

Initialize the channel correlation coefficient between channels l and m for targets as

$$\hat{\rho}_{\mathrm{t}j}(l,m) = \rho_0^{|l-m|},\ 0 < \rho_0 \le 1 \tag{49}$$

where ρ_0 is a constant, such as $\rho_0 = 0.99$ [16].

With k_{scnr}, the estimate for the AMF test for targets is initialized as

$$\hat{\omega} = \frac{\sigma_{\mathrm{s}}\mathbf{u}^{\mathsf{H}}\hat{\mathbf{a}}\hat{\mathbf{a}}^{\mathsf{H}}\mathbf{u} + \sigma_{\mathrm{n}}}{\sigma_{\mathrm{cn}}} \approx k_{\mathrm{scnr}}\mathbf{u}^{\mathsf{H}}\hat{\mathbf{a}}\hat{\mathbf{a}}^{\mathsf{H}}\mathbf{u}. \tag{50}$$

Next, the proposed detector is performed with the above target parameters and a high overall target detection Pfa in order to sense the potential targets as possible. Despite that the constant target parameters for each pixel under the test may mismatch with the true ones in the data, and thus, the proposed detector would suffer from target detection performance degradation, nevertheless, the relatively high global detection Pfa may make up for the reduced Pd. Inevitably, there would be many false alarms at the same time. Suppose that Q potential targets are detected in this process. The adaptive target parameters can be estimated for each potential target by using the maximum likelihood (ML) method [16], [18], where $\hat{\omega}_q$ is estimated by the mean of the AMF test, and $\hat{\varphi}_{\mathrm{t}j,q}$ and $\hat{\rho}_{\mathrm{t}j,q}$ are, respectively, estimated via (8) and (19), $q \in \{1, 2, \ldots, Q\}$.

Finally, retest the potential targets with the proposed method under the proper overall Pfa and adaptive target parameters. Since the target information fed back from the environment is more accurate compared with the fixed one, the test statistics under the hypothesis H_1 may be reliably predicted, and thus, the proposed detector is able to achieve enhanced target detection performance. Consequently, the real targets are remained, and those false alarms are to be removed. The key procedures of the cognitive detection are summarized in Algorithm 2. Note that in the autosensing stage, the global optimization to determine the optimal local detection Pfas is only conducted once as the essential target parameters are initialized as constants for any pixel. In the enhanced detection, the global optimization is reperformed for Q potential targets, separately. In total, the global optimization operates $Q + 1$ times via cognitive detection, which can greatly lower the computational complexity in practical applications.

Algorithm 2 Cognitive Detection Procedures

Auto-sensing process:
1. Estimate $\hat{\Theta}_{\mathrm{c}}$, and initialize $\hat{\Theta}_{\mathrm{t}}$ on constant parameters;
2. Implement the proposed detection with a high Pfa;
3. Estimate the target parameter sets $\hat{\Theta}_{\mathrm{t},1}, \ldots, \hat{\Theta}_{\mathrm{t},Q}$;

Enhanced detection on potential targets:
4. Reset local test statistics under the hypothesis H_1;
5. Implement the proposed detection under the proper Pfa;
6. Output final targets and parameters.

V. Simulations and Real-Data Experiments

A. Simulation Results

In this section, the theoretical statistics of the AMF detector and ATI-phase detector and their respective ROC curves are first investigated in terms of the heterogeneity degree of clutter background, SCNR output, target radial velocity, and baseline length. In the simulations, the effective look number is set to $\hat{n} = 2.1$, and two targets, i.e., T1 with $\omega_1 = 2$ and T2 with $\omega_2 = 10$, are considered. The AMF test pdfs for the homogeneous clutter background S1 and two targets are shown in Fig. 7(a), while those of two heterogeneous clutter backgrounds S2 with $\chi = 3$ and S3 with $\chi = 11$ and two targets are shown in Fig. 7(b). The ROC curves in terms of the Pd versus the Pfa are shown in Fig. 7(c). The pdfs of the ATI-phase test for clutter and targets are, respectively, shown by Fig. 8(a) and (b). In Fig. 8(a), the ML estimate of the ATI phase for the ground clutter is set to 0 rad, and a smaller channel correlation coefficient corresponds to the longer baseline case. In Fig. 8(b), the baseline length is fixed; for the targets "T1," "T2," and "T3," their radial velocities are gradually increased in the same quantity, and thus, φ_t are, respectively, set to 0.5, 1.5, and 1 rad. The ROC curves of the ATI-phase tests are shown in Fig. 8(c).

Based on the above simulation results, several observations can be obtained. The Pd of the AMF detection becomes worse under the same Pfa if the target SCNR is smaller or the clutter background is more heterogeneous. The ATI-phase detection performance does not rely on the heterogeneous degree of the clutter background and would suffer from rapid decline with the smaller ρ_{c} and φ_{t}. From Figs. 7(c) and 8(c), the only AMF detection and the only ATI-phase detection cannot acquire good target detection results under a low Pfa in the simulation conditions.

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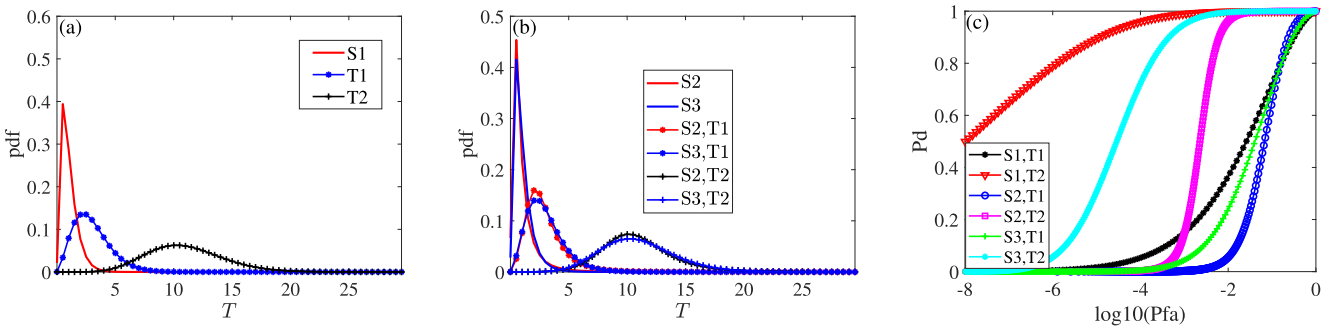


Fig. 7. Theoretical pdf comparisons for the AMF test about the backgrounds of (a) homogeneous clutter (S1) and (b) heterogeneous clutter (S2 with $\chi = 3$ and S3 with $\chi = 11$) with two targets of $\omega = 3$ dB (T1) and $\omega = 10$ dB (T2). (c) Pd versus the Pfa of the AMF detection under these statistics.

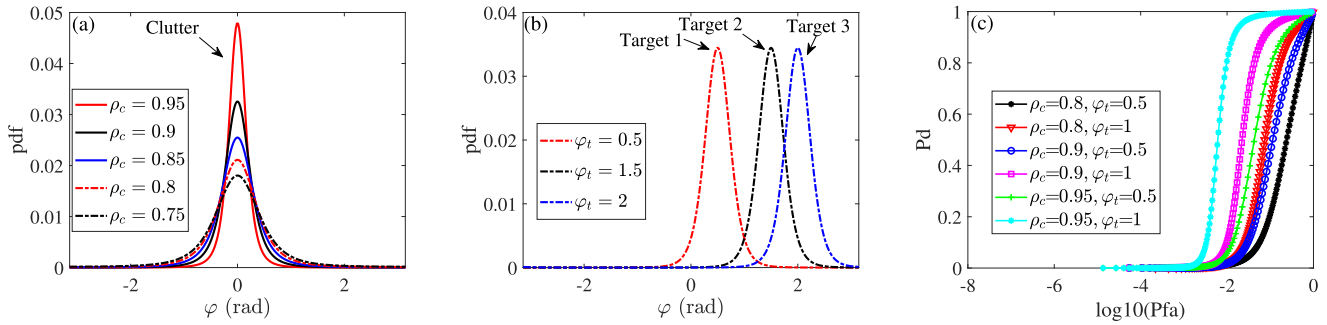


Fig. 8. Theoretical pdf comparisons for the ATI-phase tests on (a) channel correlation coefficient ρ and (b) target ATI-phase estimate φ_t. (c) Pd versus the Pfa of the ATI-phase detection under the statistics.

Then, the ROC curves for the AMF detector, ATI-phase detector, the two-step detector [16], and the proposed method are compared to illustrate the dependence of the target detection performance on the factors, i.e., heterogeneous degree of clutter background, output SCNR, target radial velocity, baseline length, and channel number.

1) ROC in Different Clutter Backgrounds and SCNRs: Fig. 9(a)–(c) illustrates the ROC curves of the compared methods in the homogeneous background, heterogeneous background with $\chi = 11$, and heterogeneous background at $\chi = 3$, respectively. The ROC curves are simulated with $N = 3$, $\hat{n} = 2.1$, $\rho(1,2) = \rho(2,3) = 0.99$, $\omega_1 = 3$ dB, $\varphi_t(1,2) = \varphi_t(2,3) = 0.5$ rad, and $\varphi_t(1,3) = 1$ rad. In Fig. 9(a)–(c), "ATI 1" and "ATI 2" are, respectively, the ATI-phase test between the first and the second channels and between the first and the third channels. In the two-step detector [16], equal local detection Pfa is adopted. For target performance comparisons in terms of SCNR output, the ROC curves with $\omega_2 = 10$ dB are shown in Fig. 10(a)–(c). From the ROC curves in Figs. 9 and 10, "ATI 2" outperforms "ATI 1" due to a longer baseline and outperforms the AMF detection for the low target SCNR case. The AMF detection performance becomes better for the target with a larger SCNR output and in a less heterogeneous background.

For the target with $\omega_2 = 10$ dB in the extremely heterogeneous background, i.e., $\chi = 3$, the two-step detection performance exceeds the ATI-phase detection and AMF detection, as shown in Fig. 10(c). However, the two-step detector cannot always have a better target detection performance compared with the ATI-phase detection and AMF detection, as shown in Fig. 9, since the fixed local detection Pfa may be not proper at some detection environments. Compared with these methods, the proposed method is able to acquire a larger Pd under the same Pfa in both homogeneous and heterogeneous backgrounds and for the SCNR outputs.

2) ROC Versus Channel Number: The target detection performance of the proposed method versus channel number (spatial DoF of the radar system) is investigated under the heterogeneous background with $\chi = 3$. In the simulation, the target SNR out of the array element channel is 10 dB. Since the adaptive clutter suppression output SCNR increases with the channel number j as the relationship $10\log 10(j)$ [1], $j = 2, 3, \ldots, 8$, the AMF test parameter ω is set as $\omega = (10 + 10\log 10(j))$ dB. As the ATI-phase test with a longer baseline can perform better, as shown in Figs. 9 and 10, the ATI-phase test with the maximum baseline length is used for the given channel number j, and the ML estimate of the target ATI phase may be reasonably set as $\varphi_t(1, j) = 0.5 \times (j-1)/2$ rad according to (9). The channel correlation coefficient follows the expression (49) at $\rho_0 = 0.99$.

The Pd versus the Pfa and the channel number for the AMF detection are shown in Fig. 11(a), while that for the ATI-phase detection is shown in Fig. 11(b). The target detection performance of the AMF detection significantly increases with the more channel number. As to the ATI-phase detection, the more channel number can provide a positive effect on target detection, while the smaller channel correlation coefficient due to the longer baseline length would cause a target detection performance degradation at the same time. Under the comprehensive influence, the target detection performance, finally, shows a few increases with more channels, as shown in Fig. 11(b). The ROC curves of the compared methods in terms

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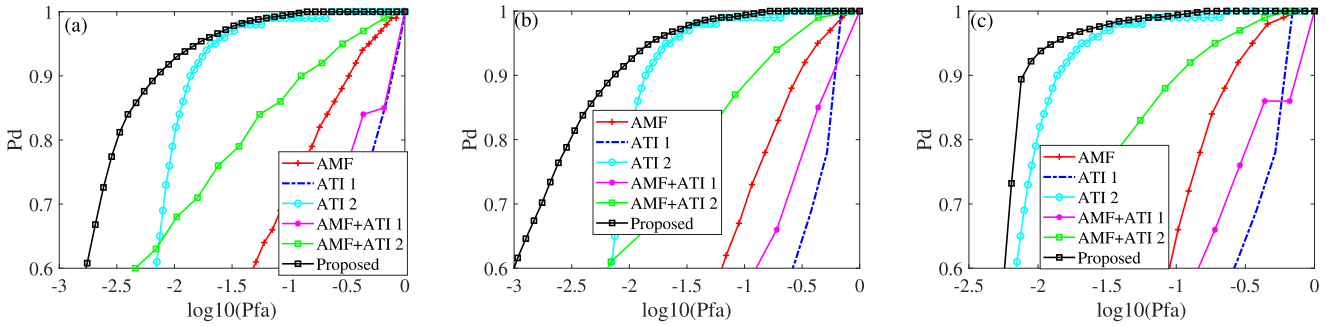


Fig. 9. ROC curves of the compared methods for (a) homogeneous background, (b) heterogeneous background (χ=11), and (c) heterogeneous background (χ=3), where target parameters are $\omega_1 = 3$ dB, $\varphi_t(1,2) = \varphi_t(2,3) = 0.5$ rad, and $\varphi_t(1,3) = 1$ rad.

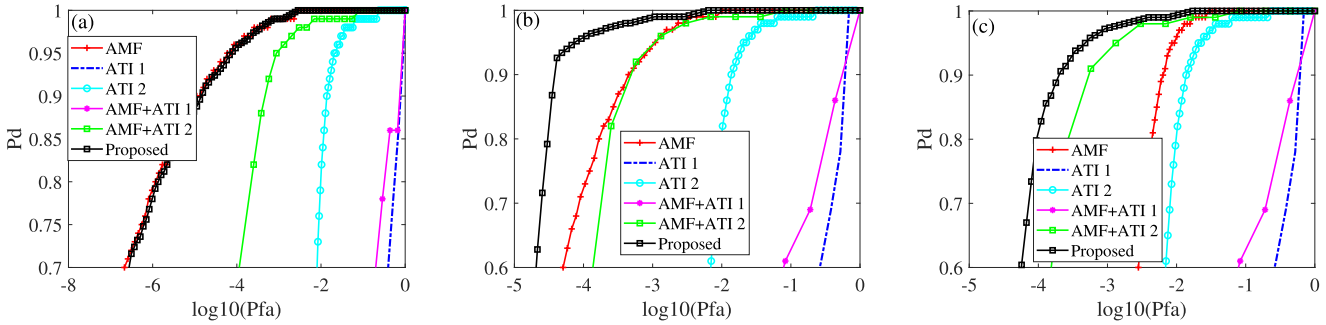


Fig. 10. ROC curves of the compared methods for (a) homogeneous background, (b) heterogeneous background (χ=11), and (c) heterogeneous background (χ=3), where target parameters are $\omega_2 = 10$ dB, $\varphi_t(1,2) = \varphi_t(2,3) = 0.5$ rad, and $\varphi_t(1,3) = 1$ rad.

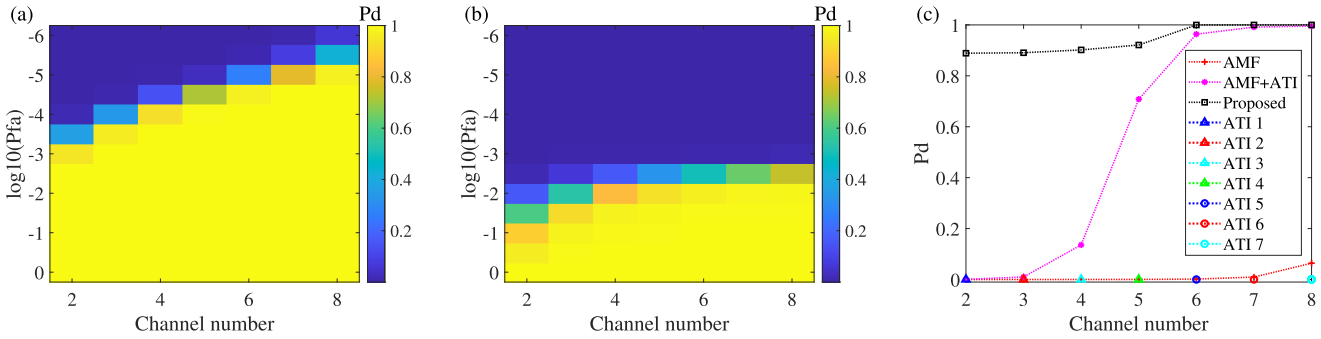


Fig. 11. Pd versus the Pfa and the channel number for (a) AMF detection and (b) ATI-phase detection. (c) ROC curves of the compared methods under the same Pfa of 10^{-6}, the heterogeneous background χ=3, and the target parameters $\omega = (10 + 10\log 10(j))$ dB and $\varphi_t(1, j) = 0.5 \times (j-1)/2$ rad for the given channel number j, $j = 2, 3, \ldots, 8$.

of the Pd versus the channel number under the Pfa 10^{-6} are shown in Fig. 11(c), where the proposed method outperforms the others and performs better with more channels.

B. Experimental Results

In this section, we investigate the target detection performance of the proposed method based on a three-channel real-world dataset, collected by the airborne SAR in the side-looking mode in China's urban areas. The compared methods include the AMF method, the ISTAP method [2], the IMP method [17], and the two-step detector [16]. For the raw data, we perform the SAR imaging, the channel coregistration, and balancing [3] and then acquire three SAR images. The main system parameters are listed in Table II.

The SAR images and the adaptive clutter suppression residuals are, respectively, shown in Fig. 12(a1) and (a2). Notice that most clutter is effectively rejected, while some large clutter residuals or potential targets appear at high values in Fig. 12(a2). Consider that the vehicle size may be 2.8 × 5 m in practice, and the corresponding SAR images would occupy one to two pixels. Thus, the single look number is adopted for constructing the AMF and multibaseline ATI-phase tests. The statistical parameters are estimated as $\hat{\chi} = 11$, $\hat{\rho}_c(1,2) = 0.9975$, $\hat{\rho}_c(1,3) = 0.9876$, and $\hat{\rho}_c(2,3) = 0.9943$. The comparisons between the clutter histogram and estimates using homogeneous and heterogeneous pdfs are shown in Fig. 12(b), where the mean square errors (mses) between the histogram and estimates are displayed by the right vertical axis. The results indicate that the pdf based on the heterogeneous clutter model [i.e., (4), (5), and (13)] can predict the real data statistics for the AMF test accurately. The comparisons between the histogram and estimate statistics for ATI-phase tests $\varphi_1 = \varphi(1,2)$, $\varphi_2 = \varphi(1,3)$, and $\varphi_3 = \varphi(2,3)$ are,

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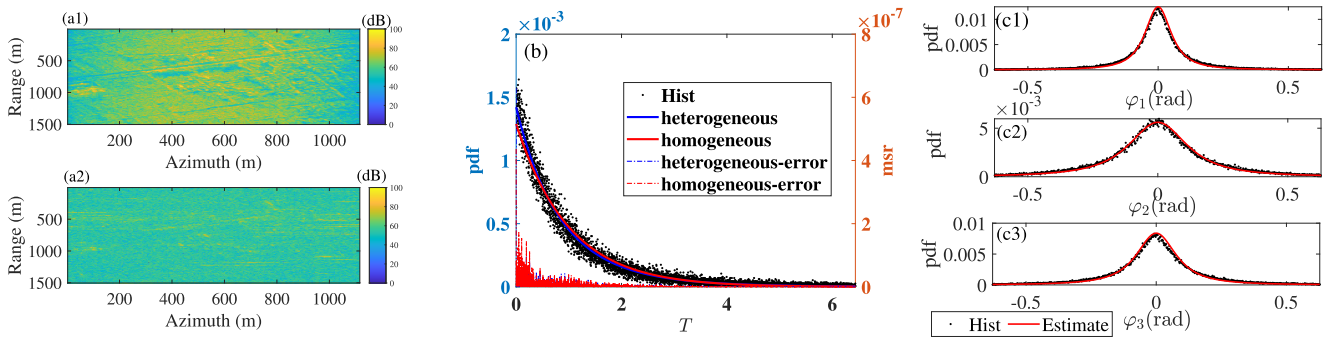


Fig. 12. Real-world data analyses. (a1) and (a2) Azimuth-range images and the adaptive clutter suppression residuals. (b) Comparisons between the histogram and estimates with homogeneous and heterogeneous statistics for the AMF test and (c1)–(c3) histogram and the estimate pdfs for three ATI-phase tests φ_1, φ_2, and φ_3.

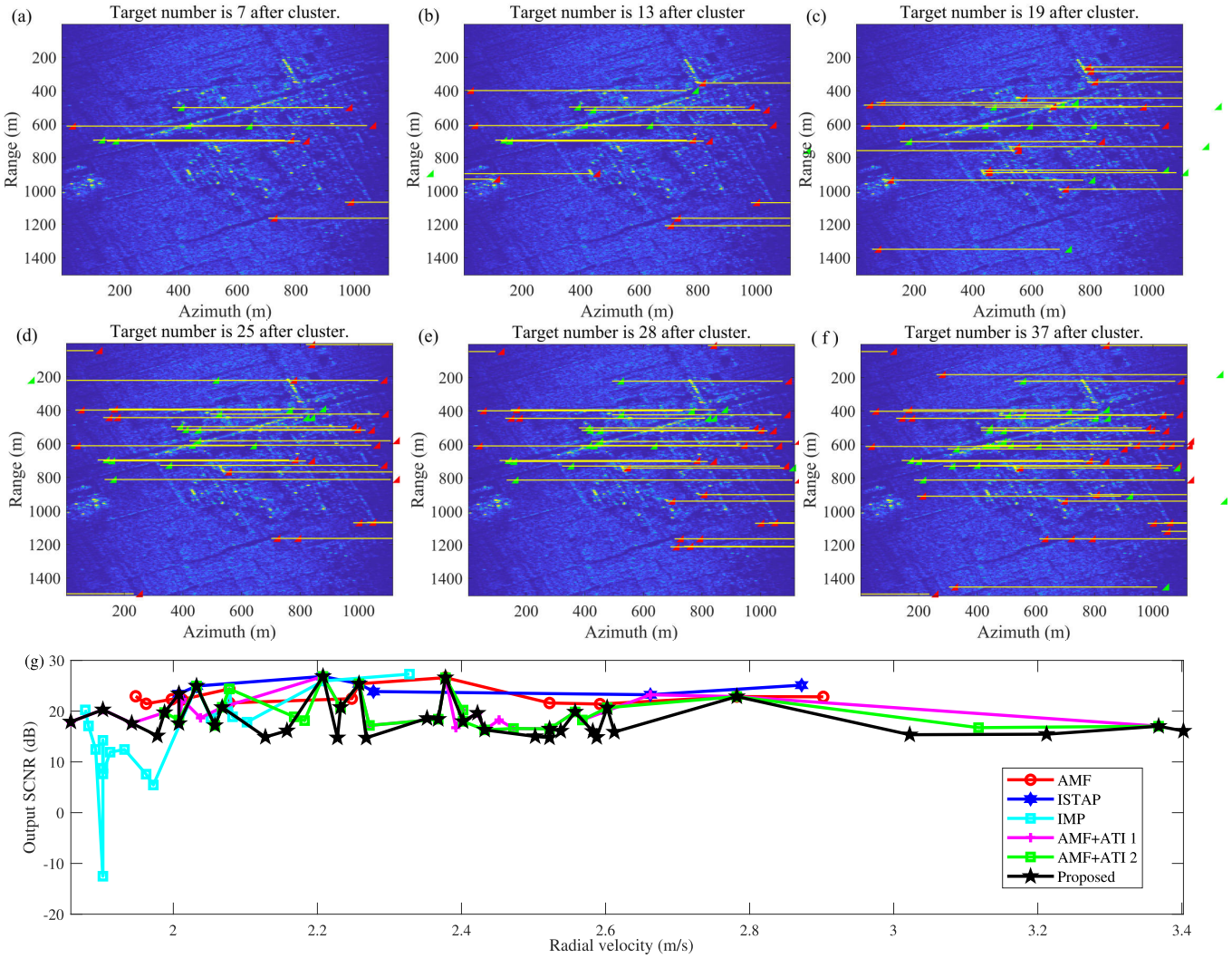


Fig. 13. Real-data target detection and relocation results under $P_f = 10^{-6}$ for (a) AMF, (b) ISTAP, (c) IMP, (d) AMF+ATI 1, (e) AMF+ATI 2, and (f) proposed method. (g) Estimation results on the potential target output SCNR and radial velocity.

respectively, shown in Fig. 12(c1)–(c3). Theoretical models properly suit the real data statistics.

Based on the statistics estimates, the compared methods are performed with the same Pfa 10^{-6}. Note that the local detection Pfa of the two-step detectors is, respectively, set as 10^{-4} and 10^{-2}; "ATI 1" and "ATI 2" are, respectively, the ATI-phase tests measured between the channels 1 and 2, and between the channels 1 and 3. For the proposed detector, the autosensing process is first conducted with the constant target parameters, $v_m = 0.5$ m/s and $k_{scnr} = 10$ dB, and a high overall Pfa 10^{-3}. In this stage, we acquire 336 potential target pixels, which are clustered based on the pixel spatial space and ATI

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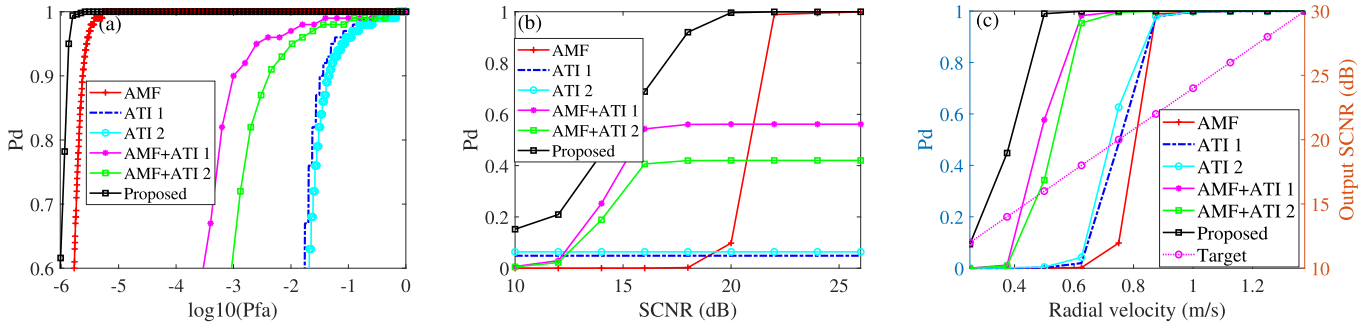


Fig. 14. ROC curves of the compared methods on the real-world clutter background in terms of Pd versus (a) Pfa at SCNR = 20 dB, (b) SCNRs under $P_{\mathrm{f}} = 10^{-6}$, and (c) target radial velocities under $P_{\mathrm{f}} = 10^{-6}$.

TABLE II
SYSTEM PARAMETERS

Name	Value	Name	Value
λ	0.02 m	N	3
Baseline	0.225 m	V	120 m/s
Nearest slant range	33 km	Resolution	5m × 5m

phase to form the potential targets. Then, their parameters are estimated, and the enhanced detection on the potential targets with the Pfa 10^{-6} is implemented. Since the moving target images would have position shifts along the azimuth direction due to their nonzero radial velocities in the SAR imaging process, we further relocate the detected targets in the SAR images relying on the relationship between the target radial velocity v_{r} and the azimuthal displacement x, $\triangle x = v_{\mathrm{r}} \times R/V$, where R denotes the target slant range. Finally, the target detection and azimuth relocation results for the compared methods are shown in Fig. 13(a)–(f), where the detected targets are marked with the red triangles, and the relocated ones are marked by the green triangles. Note that some potential targets after the pixel clustering may be on the edge of the SAR image, and some potential targets may be relocated outside the SAR image.

For the real data, as the prior knowledge of the ground moving target positions in the SAR image is unknown, the potential targets, which are detected and relocated on the road or nearby the road (considering the relocation errors), may be the true targets. In Fig. 13(a)–(f), the potential targets are, respectively, 7, 13, 19, 25, 28, and 37 for the AMF, the ISTAP, the IMP, AMF+ATI 1, AMF+ATI 2, and the proposed method. The output SCNRs and radial velocities of the potential targets are estimated, and the results are shown in Fig. 13(g). Under the same Pfa, the proposed method successfully detects more potential targets with a smaller radial velocity and a lower output SCNR, and most of the potential targets are relocated near the road.

Finally, the ROC curves in the real-data clutter background are predicted by adding simulation targets. The target detection performance of the compared methods is shown in Fig. 14, where Fig. 14(a) is the Pd versus the Pfa with $\omega = 20$ dB, and Fig. 14(b) is the Pd versus the output SCNR under $P_{\mathrm{f}} = 10^{-6}$, while Fig. 14(c) is the Pd versus the target radial velocities under $P_{\mathrm{f}} = 10^{-6}$. In Fig. 14(a) and (b), $\varphi_{\mathrm{t}}(1,2) = \varphi_{\mathrm{t}}(2,3) = 0.419$ rad, and $\varphi_{\mathrm{t}}(1,3) = 0.846$ rad. In Fig. 14(c), the radial velocities of the simulation targets are shown by the horizontal axis, and the output SCNRs are shown by the right vertical axis. It can be observed that the proposed method can acquire the better Pd under the same Pfa, as shown in Fig. 14(a), and detect the target with a smaller SCNR, as shown in Fig. 14(b), and a smaller radial velocity, as shown in Fig. 14(c). Thus, the results can reliably validate the effectiveness of the proposed method in practical applications and verify the significant advantages of GMTI over the compared methods.

VI. CONCLUSION

We propose the optimal fusion-based target detection with multichannel ATI SAR for GMTI in the heterogeneous background. In the product-model heterogeneous clutter plus Gaussian noise background, the maximum target detection performance based on the multibaseline ATI-phase tests and the AMF test is obtained by using the optimal local detection thresholds in terms of Chair and Varshney's optimal fusion rule via global optimization. For the system channel number $N \gg 2$, a low-complexity solution to the global optimization processing is given by assuming equal local detection threshold for the independent and identically distributed (i.i.d.) ATI-phase tests. For practical implementation, the cognitive detection framework for the proposed detector is given to achieve target detection without prior knowledge.

Based on the simulation analyses, the AMF test and the standard ATI-phase test, respectively, show advantages in dealing with the low SCNR target and the heterogeneous clutter but cannot perform properly for both cases. Although the two-step detector by cascading the AMF test and the standard ATI-phase test can improve the target detection performance but would suffer from performance degradation under the severe conditions that more heterogeneous clutter, less SCNR, and slower targets appear, the proposed detector shows an improved target detection performance for the varied clutter backgrounds and different targets over the aforesaid methods. Besides, the target detection results and the metrics of the ROCs on real-data experiments effectively validate the superiority of the proposed method against the mainstream methods in practical applications.

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Min Tian (Member, IEEE) received the B.Eng. degree from Nanchang Hangkong University, Nanchang, China, in 2014, and the Ph.D. degree from Xidian University, Xi'an, China, in 2019.

From 2019 to 2022, she was a Research Assistant with Xi'an Institution of Space Radio Technology, Xi'an. From 2022 to 2024, she was a Research Assistant with the Research Institute of Xidian University, Hangzhou, China, where she is currently an Associate Researcher. Her research interests include radar clutter simulation and characterization, adaptive signal processing, and detection.



Bin Liao (Senior Member, IEEE) received the B.Eng. and M.Eng. degrees from Xidian University, Xi'an, China, in 2006 and 2009, respectively, and the Ph.D. degree from The University of Hong Kong, Hong Kong, in 2013.

From 2013 to 2014, he was a Research Assistant with the Department of Electrical and Electronic Engineering, The University of Hong Kong, where he was also a Research Scientist with the Department of Electrical and Electronic Engineering from August 2016 to October 2016. From March 2023 to June 2023, he was a Visiting Professor with the Centre for Wireless Innovation (CWI), Queen's University Belfast, Belfast, U.K. He is currently a Full Professor with the College of Electronics and Information Engineering, Shenzhen University, Shenzhen, China. His research interests include sensor array processing, adaptive filtering, and convex optimization, with applications to radar, navigation, and communications.

Dr. Liao was a recipient of the Best Paper Award in 2016 21st International Conference on Digital Signal Processing and the 2017 22nd International Conference on Digital Signal Processing. He was a recipient of the National Natural Science Foundation of China and European Research Council (NSFC-ERC) Programme. He is an Associate Editor of IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS and *Multidimensional Systems and Signal Processing*.

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